

La característica de un cuerpo es 0 o un número primo

Lema 1. Sea K un cuerpo, entonces $\text{char}(K) = 0$ o $\text{char}(K) = p$ con p primo.

Demostración: Supongamos que $\text{char}(K) = mn$ con $m, n \in \mathbb{N}_{>1}$. Entonces,

$$(1 + 1 + \dots + 1) \cdot (1 + 1 + \dots + 1) = 1 + 1 + \dots + 1 = 0$$

Como en K no hay divisores de 0, entonces $1 + 1 + \dots + 1 = 0$ o $1 + 1 + \dots + 1 = 0$.

Como $n < mn$ y $m < mn$, entonces $\text{char}(K) = m$ o $\text{char}(K) = n$ y $\text{char}(K) < mn$ lo que contradice la definición de característica. Por tanto, $\text{char}(K)$ no puede ser compuesto. ■

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